

Riddle Registry

 **Security Warning:** Work in an isolated environment (virtual machine or container). Never open unknown files on your primary machine.

Challenge Description

Hi, **intrepid investigator**!  You've stumbled upon a peculiar PDF filled with what seems like nothing more than garbled nonsense. But beware! Not everything is as it appears. Amidst the chaos lies a hidden treasure—an elusive flag waiting to be uncovered.

Hints

- Don't be fooled by the visible text; it's just a decoy!
 - Look beyond the surface for hidden clues
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Tools Used

- **file:** Utility to identify file types
 - **strings:** Command-line tool to extract readable ASCII sequences from files
 - **grep:** Text search utility for filtering output
 - **exiftool:** Metadata inspection tool
 - **base64:** Encoding/decoding utility for Base64 data
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Complete Solution

Step 1: Download and Prepare

Download the PDF file named `confidential.pdf` from the challenge page.

Title: *The Ultimate Guide to Flag Hunting*

Welcome to the challenge!

Don't worry, this might look like gibberish, but maybe there's something hidden somewhere? I spent so much time creating this PDF with care... or maybe not! 😊

Here's a Quick Story:

Once upon a time, in a land far, far away, there was a secret... But where could it be? Hidden deep within the document? Maybe the text holds clues?

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Integer posuere erat a ante venenatis dapibus posuere velit aliquet. [REDACTED] Fusce dapibus, tellus ac cursus commodo, tortor mauris condimentum nibh, ut fermentum massa justo sit amet risus. Curabitur blandit tempus porttitor. [REDACTED], consectetur adipiscing elit.

You thought this was important? Nah, it's just random text. Keep looking. Or maybe, just maybe, you're in the wrong place?

Special Hidden Section:

[REDACTED]

Don't bother trying to reveal the hidden text, it's just nonsense anyway. Even if you somehow manage to do it, all you'll get is:

[REDACTED]

If you're still reading this, I'll tell you a secret: the answer might not be here after all... 😊

Good luck! You'll need it! 🍀

Step 2: Basic File Inspection

First, verify the file type and general information:

```
file confidential.pdf
```

Expected output:

```
confidential.pdf: PDF document, version 1.7, 1 page(s)
```

Step 3: Search for Visible Text

The flag may be present as plain text. Try extracting all character strings:

```
strings confidential.pdf | grep -i "picoCTF"
```

⚠ *If no results appear, proceed to the next step (metadata).*

Step 4: Inspect Metadata

This is the crucial step! Use `exiftool` to examine all PDF metadata:

```
exiftool confidential.pdf
```

Complete output:

```
ExifTool Version Number      : 13.50
File Name                    : confidential.pdf
Directory                   : .
File Size                    : 183 kB
File Modification Date/Time   : 2025:11:07 21:17:38+02:00
File Access Date/Time        : 2026:05:04 16:43:55+02:00
File Inode Change Date/Time   : 2026:05:04 16:43:48+02:00
File Permissions              : -rw-r--r--
File Type                    : PDF
File Type Extension          : pdf
MIME Type                    : application/pdf
PDF Version                  : 1.7
Linearized                   : No
Page Count                   : 1
Producer                     : PyPDF2
Author                       :
cGljb0NURntwdXp6bDNkX20zdGFkYXRhX2YwdW5kIV80MjQ0MGM3ZH0=
```

🔑 **Key point:** Notice the `Author` field which contains a suspicious, encoded string!

Step 5: Decode the Base64 Value

The string in the `Author` field is encoded in Base64:

```
cGljb0NURntwdXp6bDNkX20zdGFkYXRhX2YwdW5kIV8zNTc4Nm5YX0=
```

Decode it using one of these commands:

```
echo 'cGljb0NURntwdXp6bDNkX20zdGFkYXRhX2YwdW5kIV8zNTc4NzM5YX0=' | base64 --decode
```

or

```
echo 'cGljb0NURntwdXp6bDNkX20zdGFkYXRhX2YwdW5kIV8zNTc4NzM5YX0=' | base64 -d
```

✅ Final Result

picoCTF{<REDACTED>}

🧠 Key Concepts

- **Metadata:** Hidden information in files (author, date, producer, etc.)
- **Base64:** Common encoding to obscure or compress content
- **Anomaly Recognition:** Looking for elements that seem suspicious or out of place